

Ashok Gahatraj

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Education

M.S. in Applied Geography | New Mexico State University, Las Cruces, NM | 2024- Present

- Thesis: *Remote Sensing of Vegetation Phenology in Southwestern U.S. Sky Islands: A Comparative Analysis of the Sonoran, Chihuahuan, and Mojave Deserts*
- Advisor: Dr. Michaela Buenemann

B.E. in Geomatics Engineering | Tribhuvan University, Pokhara, Nepal | 2017-2022

Research Experience

Graduate Research – Vegetation Phenology in Sky Islands | 2024 – Present

- Analyzing vegetation phenology dynamics across the Sonoran, Chihuahuan, and Mojave Deserts using MODIS and Landsat time-series data.
- Tools: Phenofit, BFAST, R, GEE, LandTrendr, ArcGIS

Research Assistant – ArcGIS Indoors for Research Space Management | Summer 2025

- Designed and deployed an ArcGIS Indoors Information Model for the NMSU, including floor plan integration and data schema design.
- Created interactive ArcGIS Dashboards and a pilot ArcGIS Hub site to communicate research space insights with stakeholders.
- Tools: ArcGIS Pro, ArcGIS Online (Dashboards, Indoors, Experience Builder, Hub), R

Urban Growth and Environmental Footprint Analysis | 2024 – 2025

- Investigating the relationship between urban expansion, vegetation change, and land surface temperature in the Northern Rio Grande Corridor using MODIS time-series data.

Undergraduate Research – Waste Management System | 2022

- Designed a GIS-based solid waste management plan for Pokhara Metropolitan City.

Flood Prediction Model Using Early Warning System Messages | Rapid Prototype Design Hackathon | 2021

- Developed a flood prediction prototype integrating SMS-based early warning systems and geospatial visualization.

Publications

Gahatraj, A., Bhusal, K. (2025). *Assessing the Relationship Between Land Use/Land Cover Change, NDVI, NDBI, and Land Surface Temperature: A 20-Year Analysis of Rupandehi District, Nepal*. Doi: [10.4236/oalib.1114123](https://doi.org/10.4236/oalib.1114123)

Conferences & Presentations

Poster: *Mapping and Managing Research Space: An ArcGIS Indoors Pilot at New Mexico State University*—SWAAG Conference 2025

Attendee: Esri User Conference, 2025, San Diego, CA

Poster: *Urban Growth and Land Surface Temperature in the Northern Rio Grande Corridor* – AAG Conference 2025

Poster: *Urban Growth and Land Surface Temperature in the Northern Rio Grande Corridor* – SWAAG Conference 2024

Poster: *Integrating Remote Sensing & GIS for Waste Management Planning in Pokhara Metropolitan City* – Geomatics Engineering Association of Nepal 2021

Video Presentation: *SMS-Based Flood Prediction System for Early Warning* – Rapid Prototype Design Challenge 2021

Awards & Honors

- 3rd Place in poster presentation in SWAAG, 2025
- Robert and Elizabeth Czerniak Endowed Scholarship (2025)
- Richard D. Wright Graduate Award for Excellence in Applied GIS (2025)

Teaching Experience

Graduate Teaching Assistant – Department of Geography & Environmental Studies, NMSU | Aug 2024 – Present

- Assisted with undergraduate lab instruction in Physical Geography and GIS.
- Supported students with lab exercises, grading, and geospatial analysis software training.

Work Experience

- Data Analyst – Numeric Mind, Kathmandu, Nepal | Oct 2022 – Jun 2024
- Data Quality Intern – Humanitarian OpenStreetMap Team (HOT OSM) | Aug 2020 – Oct 2020

Certifications & Training Selected.

- FAA Part 107 Certified Drone Pilot (2025)
- Introduction to GIS Programming (2025, online course)
- Google Data Analytics Specialization (Coursera)
- Data Analyst with R (Datacamp)
- Geospatial Analysis with R (Udemy)
- Astronomy: Exploring Time and Space-ASU (Coursera)
- Data Analysis with Python (Coursera)
- Mastering Earth Observation with AI & Google Earth Engine (Udemy)

Skills

- Programming & Analytics: Python, R, SQL, SAS, Excel
- Remote Sensing & GIS: ArcGIS Pro, ArcGIS Indoors, QGIS, ENVI, Google Earth Engine, Agisoft
- Data Visualization: Power BI, Matplotlib, ggplot2, Rasterio
- Surveying & UAV: Total Station, DGPS, UAV (FAA Part 107 Certified)
- Languages: English, Nepali, Basic Spanish

Volunteering

- Conference Assistant: SWAAG Annual Meeting, 2025
- Conference Assistant: AAG Annual Meeting, 2025
- Volunteer: Educators' Job Fair, New Mexico State University, 2025
- IT & Communication Volunteer: Nepalese Student Association (NeSA) 16th Conference, 2025

Professional Affiliations

- Graduate GES Organization – Treasurer(2025-Presesnt)
- American Association of Geographers (AAG) – Member (2024–Present)
- Gamma Theta Upsilon – Member (2025–Present)
- Nepalese Student Association (NeSA) – Member (2024–Present)
- Nepal Engineering Council (NEC) – Member (2022–Present)